

Appendix A: Parameter definition

Table A.1: Overview of parameter definition for the circularity indicator

Symbol	Definition
M	Mass of material in product
V	Virgin material
F_u	Fraction of reused components
F_r	Fraction of recycled content of the produced feedstock
F_{rc}	Fraction of recycled content of a product
R_{in}	Recycled material used as input for feedstock production
R_{fp}	Scrap generated during feedstock production
R_{cp}	Scrap generated during component production
R_{EoL}	Recycled material recovered and recycled at end-of-life (EoL)
R_{out}	Total amount of recycled material recycled ($R_{fp} + R_{cp} + R_{EoL}$)
R	Recycled material
C	Reused component
W_{fp}	Waste from feedstock production
E_{fp}	Efficiency of feedstock production
C_{fp}	Fraction of material losses recovered as scrap during feedstock production
W_{cp}	Waste from component production
E_{cp}	Efficiency of component production
C_{cp}	Fraction of material losses recovered as scrap during component production
W_u	Uncollected EoL product
C_u	Fraction of product at end of use collected for reuse of component
C_r	Fraction of product at end of use collected for material recycling
W_{ms}	Waste from material separation
E_{ms}	Efficiency of material separation process
W_{rfp}	Waste from recycled feedstock production
E_{rfp}	Efficiency of recycled feedstock production
W	Unrecoverable waste
X	Utility factor
L	Period of time product is in use (expressed in years)
L_d	Design life of a product (expressed in years) based on average market expectation
I	Use intensity during use phase
I_d	Assumed use intensity during design stage based on average consumer behaviour
$FUDC$	Available or used functional usage duty cycles
$FUDC_d$	Expected functional units based on general design for specific product category
LFI	Linear Flow Index